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Iot Based Smart Home Monitoring and Intelligent Decision System

Experiential Learning

REPORT

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In partial fulfilment for the award of degree of

Bachelor of Engineering

in

Department of Information Science and Engineering

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CERTIFICATE

Certified that the project work titled '*IoT Based Smart Home Monitoring and Intelligent Decision System*' is carried out by **Aayushi Priya (1RV24IS005)**, **Abhijay M S (1RV24IS006)**, **Bellamkonda Likhith Praveen (1RV24IS024)**, and **Lakshmi G Ravishankar (1RV24IS063)** who are bonafide students of RV College of Engineering, Bengaluru, in partial fulfilment for the award of degree of **Bachelor of Engineering in Information Science and Engineering** of the **Visvesvaraya Technological University**, Belagavi during the academic year 2025-2026. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the report deposited in the departmental library. The report has been approved as it satisfies the academic requirements in respect of experiential learning work prescribed by the institution for the said degree.

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DECLARATION

We, **Aayushi Priya, Abhijay M S, Bellamkonda Likhith Praveen, Lakshmi G Ravishankar**, students of Fourth semester B.E., department of Information Science Engineering, RV College of Engineering, Bengaluru, hereby declare that Experiential Learning (Lab) titled '***IoT Based Smart Home Monitoring and Intelligent Decision System***' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering in Information Science and Engineering** during the academic year 2025-26

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering®, Bengaluru and we will be one of the authors of the same.

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ABSTRACT

The rapid advancement of the Internet of Things (IoT) has led to the development of smart home systems capable of improving environmental monitoring and residential safety. Existing smart home solutions primarily rely on individual sensor thresholds and simple alert mechanisms, which often fail to capture complex scenarios involving multiple environmental parameters. These approaches provide limited situational awareness and lack intelligent decision-making capabilities. To address these limitations, this work proposes an IoT-Based Smart Home Monitoring and Intelligent Decision System capable of continuously monitoring temperature, humidity, smoke concentration, motion activity, and ambient light intensity. The primary objective of the proposed system is to enhance home awareness and safety by integrating sensor fusion, state-based analysis, and graph-theoretic concepts with cloud-enabled monitoring.

The proposed system consists of multiple sensors interfaced with an ESP32 controller that performs data acquisition and cloud communication. Sensor readings are transmitted to the ThingSpeak platform through Wi-Fi and MQTT protocols for real-time monitoring and historical visualization. A sensor fusion layer aggregates information from multiple sensors and performs preprocessing before passing the data to a Finite State Machine (FSM), which classifies the system into Normal, Warning, and Critical states. Graph theory concepts are employed to model the smart home network as a directed graph and perform connectivity and shortest-path analysis. The system was implemented using ESP32, DHT11, MQ-2, PIR, and LDR sensors, while ThingSpeak was used for cloud visualization. The proposed approach is applicable to smart homes, building automation systems, and intelligent safety monitoring applications.

Experimental observations demonstrated that the developed system was able to continuously monitor environmental parameters and generate alerts whenever abnormal conditions were detected. The Finite State Machine successfully classified system conditions into different operational states, and graph-based analysis effectively represented the communication structure of the smart home network. Real-time visualization and historical monitoring were achieved through the ThingSpeak dashboard. Compared to conventional threshold-based monitoring systems reported in the literature, the proposed approach provides improved situational awareness by considering multiple sensor readings collectively and incorporating state-based decision-making mechanisms, thereby reducing the limitations associated with independent sensor analysis.

In conclusion, the proposed IoT-Based Smart Home Monitoring and Intelligent Decision System successfully integrates sensing, communication, cloud monitoring, and intelligent analysis to provide enhanced home safety and environmental awareness. The modular architecture adopted in the system improves scalability and facilitates future enhancements. The integration of sensor fusion, finite state machine concepts, and graph-theoretic analysis represents an interdisciplinary approach that combines IoT with discrete mathematics and intelligent decision-making. Future work may include the incorporation of machine learning techniques for predictive hazard detection, support for additional sensors, development of mobile applications, and deployment in larger smart buildings and industrial environments.

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Chapter 1: Introduction

1.1 Introduction

The rapid growth of the Internet of Things (IoT) has enabled the development of smart home systems capable of monitoring environmental conditions and improving occupant safety. Most conventional smart home systems rely on independent threshold-based mechanisms, where each sensor is evaluated separately and alarms are triggered whenever predefined limits are exceeded. Although such approaches are simple to implement, they often fail to capture complex situations involving multiple interacting factors and may lead to false alarms or missed hazards.

To overcome these limitations, this project proposes an IoT-Based Smart Home Monitoring and Intelligent Decision System. The system employs multiple ESP32 sensor nodes to continuously monitor temperature, humidity, gas concentration, motion, and ambient light. Sensor readings are transmitted using MQTT to a Raspberry Pi-based edge device, where sensor fusion techniques are used to combine information from different sensors and generate an overall risk score. A Finite State Machine (FSM) is then used to determine the current operating state of the system and trigger appropriate actions. In addition, graph-based algorithms are utilized to analyze the environment and provide intelligent evacuation path recommendations. The processed information is visualized through a real-time dashboard and stored for historical analysis and cloud monitoring.

1.2 Problem Statement

The objective of this project is to develop an IoT-Based Smart Home Monitoring and Intelligent Decision System capable of continuously monitoring environmental conditions, assessing potential hazards, and providing intelligent responses using real-time sensor data. The proposed system integrates multiple ESP32 sensor nodes with a Raspberry Pi-based edge computing platform to collect, process, and analyze information from various sensors including temperature, humidity, gas concentration, motion, and ambient light.

Conventional smart home monitoring systems primarily rely on fixed threshold-based mechanisms, where each sensor is evaluated independently and alerts are generated whenever individual sensor values exceed predefined limits. Although such approaches are simple and widely adopted, they are unable to effectively capture complex situations involving multiple environmental factors. This isolated decision-making may result in false alarms, missed detections, and limited awareness of the overall condition of the environment.

Furthermore, traditional systems provide only basic warning mechanisms and lack intelligent decision-making capabilities. They do not consider the relationships among different sensor readings or represent the varying operational states of the system during abnormal conditions.

Therefore, there is a need for an intelligent monitoring framework capable of integrating information from multiple sensors, evaluating the overall risk associated with the environment, and providing meaningful insights through state-based analysis and real-time visualization. The proposed system addresses these challenges by employing sensor fusion techniques, finite state machine-based state analysis, graph-based algorithms, and a web-based dashboard to enhance the reliability and intelligence of smart home safety systems.

1.3 Objectives

1. To develop an IoT-enabled smart home monitoring system capable of continuously monitoring environmental and safety parameters such as temperature, humidity, smoke, motion, and ambient light.
2. To establish reliable communication between sensor nodes and the cloud through MQTT and ThingSpeak for real-time monitoring and data visualization.
3. To intelligently analyze sensor readings using sensor fusion and state-based decision-making techniques to provide a better understanding of the overall condition of the home.
4. To model the smart home environment using graph-based approaches and apply suitable algorithms for connectivity analysis and evacuation path determination during hazardous situations.
5. To provide real-time alerts and a user-friendly dashboard for improved awareness, safety, and monitoring of the home environment.

1.4 Significance of the Project

The proposed system enhances conventional smart home monitoring by introducing intelligent analysis and decision-making capabilities in addition to basic sensing and alert mechanisms. By combining information from multiple heterogeneous sensors, the system is capable of obtaining a more comprehensive understanding of the surrounding environment and improving the reliability of hazard detection.

The use of sensor fusion enables multiple environmental parameters to be considered collectively rather than independently, thereby reducing the limitations of simple threshold-based monitoring. Finite State Machine (FSM) concepts facilitate structured state-based decision-making, while graph-theoretic approaches allow the smart home environment to be modeled as a network for communication analysis and evacuation path determination. The integration of MQTT communication, cloud connectivity through ThingSpeak, and a real-time dashboard provides continuous monitoring and accessibility of sensor information.

In addition to its practical applications in residential safety, the proposed architecture demonstrates the integration of concepts from Internet of Things, Computer Networks, Data Structures and Algorithms, Discrete Mathematics, and Edge Computing. Owing to its modular design and scalability, the system can be extended to larger smart homes, intelligent buildings, and industrial safety monitoring applications.

Chapter 2: Solution Design

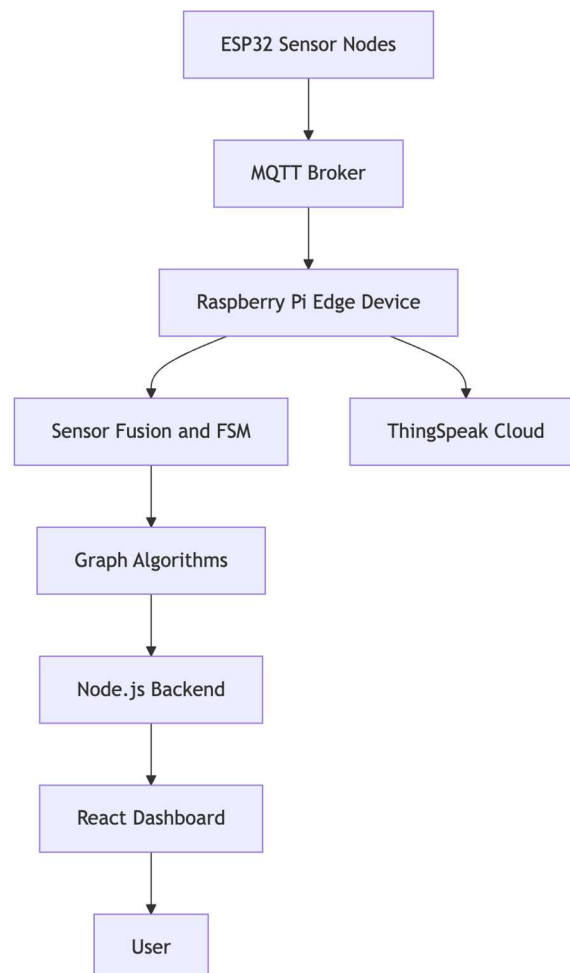
2.1 Overall System Design

The proposed IoT-enabled Smart Home Monitoring and Intelligent Decision System adopts a layered architecture consisting of sensing, communication, edge processing, and visualization components. The system is designed to continuously monitor environmental and safety parameters and provide intelligent analysis of the collected data.

Three ESP32 nodes are deployed to acquire sensor information from different parts of the house. The environmental node measures temperature, humidity, and gas concentration using DHT11 and MQ-2 sensors. The motion node employs a PIR sensor to detect movement and controls the buzzer during emergency conditions. The light node measures ambient light intensity using an LDR and controls LEDs for visual indication.

Sensor readings are transmitted through the MQTT protocol to a Raspberry Pi, which acts as the edge processing unit. The Raspberry Pi executes several algorithms including sensor fusion, state analysis using a Finite State Machine (FSM), health monitoring, and graph-based algorithms for path analysis. The processed data is then sent to a Node.js backend, which communicates with a React-based dashboard through WebSocket connections. Historical sensor data is stored and uploaded to ThingSpeak for cloud-based monitoring.

2.2 System Architecture



The proposed system consists of four major layers:

1. Sensor Layer

This layer consists of three ESP32 nodes responsible for collecting environmental and safety-related data.

Node	Sensor Used	Parameters Monitored
Environmental Node	DHT11, MQ-2	Temperature, Humidity, Gas concentration
Motion Node	PIR sensor, Buzzer	Motion detection
Light Node	LDR, LED	Ambient light levels

2. Communication layer

The communication layer employs the Message Queuing Telemetry Transport (MQTT) protocol, a lightweight publish-subscribe communication protocol widely used in IoT applications due to its low bandwidth requirements and efficient message delivery.

The three ESP32 sensor nodes periodically publish sensor readings to specific MQTT topics through the MQTT broker. The Raspberry Pi acts as a subscriber and receives these messages to aggregate data from all nodes for further processing. After processing, the Raspberry Pi publishes the analyzed information to MQTT topics that are accessed by the backend server. The Node.js backend subscribes to these topics and forwards the information to the React dashboard through WebSocket connections, enabling real-time visualization.

3. Edge Processing Layer

The Raspberry Pi acts as the central processing unit of the proposed system and performs several processing tasks to transform raw sensor readings into meaningful information.

- **Sensor Calibration:** Corrects sensor readings and performs normalization to improve measurement accuracy.
- **Sensor Fusion:** Combines information from multiple sensors to generate a single risk score representing the overall condition of the environment.
- **Risk Score Generation:** Quantifies the severity of the current environmental conditions using weighted sensor values.
- **Health Monitoring:** Continuously checks sensor readings to detect abnormal behavior or sensor failures.
- **Finite State Machine (FSM):** Determines the operating state of the system based on the calculated risk score and health status.
- **Graph Engine:** Represents different zones of the smart home as a graph and dynamically updates their connectivity and weights.
- **Breadth First Search (BFS):** Traverses neighboring zones to propagate alerts across the smart home network.
- **Depth First Search (DFS):** Performs complete graph traversal to analyze connectivity among all zones.
- **Dijkstra Algorithm:** Determines the safest evacuation path by considering the risk associated with each zone.

- **Dataset Recorder:** Stores processed sensor readings and system states for historical analysis and future reference.
- **ThingSpeak Client:** Uploads sensor data and system information to the ThingSpeak cloud platform for remote monitoring and visualization.

4. Application Layer

The application layer consists of a Node.js backend server and a React-based dashboard. The backend provides REST APIs and WebSocket communication for transferring processed data to the user interface.

The dashboard offers the following functionalities:

- **Real-Time Sensor Monitoring:** Displays the latest environmental parameters received from the system.
- **Risk Score Visualization:** Provides a graphical representation of the current risk level.
- **Historical Trend Analysis:** Displays variations in sensor readings over time using charts.
- **Finite State Machine Visualization:** Displays the current operating state and state transitions of the system.
- **Graph Visualization and Evacuation Path Display:** Illustrates the connectivity between zones and highlights the path generated by graph algorithms during hazardous situations.

2.3 Hardware Design

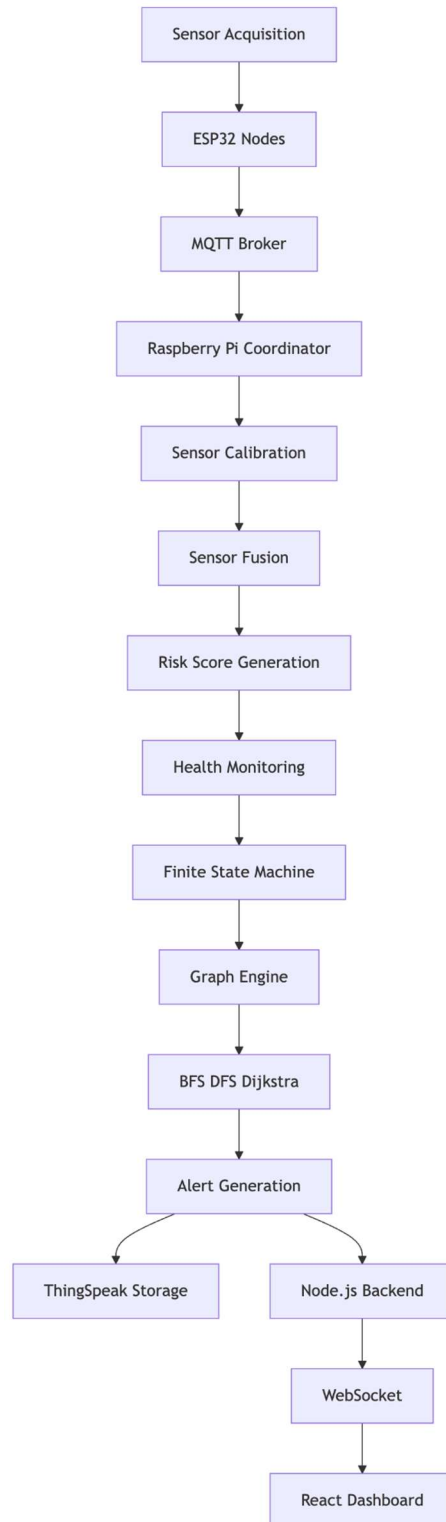
1. **Environmental Node:** The environmental node is implemented using an ESP32 microcontroller connected to: DHT11 sensor for temperature and humidity measurement and MQ-2 sensor for smoke and gas detection. The node periodically publishes sensor values to the MQTT broker.
2. **Motion Detection Node:** The motion node consists of: ESP32, PIR sensor to detect motion and active buzzer for audio alert. Whenever motion is detected or the system enters an alert state, the buzzer is activated.
3. **Light Monitoring Node:** The light monitoring node contains: ESP32, LDR sensor to measure ambient light levels and LED indicators. This node measures ambient light intensity and provides visual feedback through LED.
4. **Raspberry Pi Edge Device:** The Raspberry Pi acts as the coordinator of the entire system. It receives data from all sensor nodes and executes the intelligent decision-making algorithms.

2.4 Software Design

The software architecture of the proposed system is organized into four major modules: the ESP32 firmware, the Raspberry Pi edge coordinator, the backend server, and the frontend dashboard. Each module performs a specific function and communicates with other modules to provide real-time monitoring and intelligent decision-making capabilities.

1. **ESP32 Firmware:** The firmware is responsible for reading sensor values, creating JSON payloads and publishing data through MQTT
2. **Raspberry Pi Coordinator:** The coordinator performs sensor aggregation, data calibration, risk assessment, state transitions, graph analysis, alert generation and data storage.
3. **Backend Server:** The Node.js server performs MQTT subscription, REST API handling, historical data retrieval, and WebSocket communication with the dashboard
4. **Frontend Dashboard:** The React dashboard provides live sensor readings, risk score display, alert notifications, historical charts, Dijkstra path visualization and FSM state visualization.

2.5 Data Flow



2.6 Algorithmic Design

1. Sensor Fusion

Sensor fusion combines temperature, humidity, gas concentration, motion, and light information to generate a single risk score representing the overall condition of the environment.

This approach allows multiple sensor readings to be considered together instead of evaluating each sensor independently.

2. Finite State Machine

The Finite State Machine (FSM) models the behavior of the proposed system using six states, where transitions are determined based on the computed risk score and sensor health conditions.

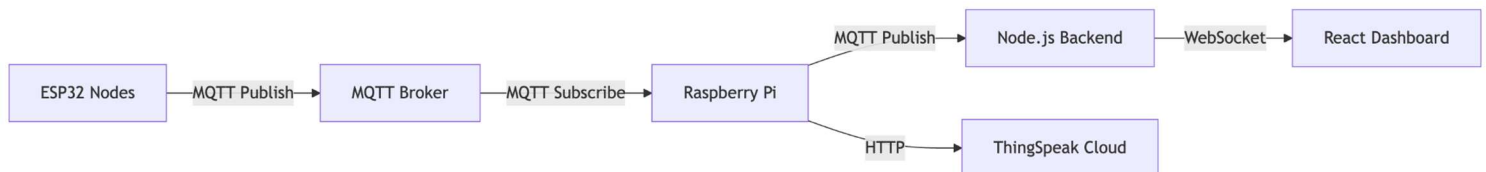
- IDLE (σ_0 , $R < 0.20$): Represents normal operating conditions in which all environmental parameters remain within safe limits and no abnormal activity is detected.
- MONITOR (σ_1 , $0.20 \leq R < 0.40$): Indicates minor variations in sensor readings that require continuous observation but do not warrant immediate action.
- ALERT_LOW (σ_2 , $0.40 \leq R < 0.65$): Represents moderately abnormal conditions where early warning notifications are generated to increase user awareness.
- ALERT_HIGH (σ_3 , $0.65 \leq R < 0.85$): Corresponds to potentially hazardous situations where the system identifies a high level of risk and triggers stronger alert mechanisms.
- CRITICAL (σ_4 , $R \geq 0.85$): Represents severe environmental conditions requiring immediate attention and emergency response measures.
- FAULT (σ_5): Indicates sensor malfunction, communication failure, or invalid readings, thereby enabling fault detection and system maintenance.

3. Graph-Based Analysis

The smart home environment is represented as a graph consisting of different zones. Graph algorithms are used for:

- Breadth First Search (BFS): To propagate alerts across connected zones.
- Depth First Search (DFS): To traverse all nodes and analyze connectivity.
- Dijkstra Algorithm: To determine the safest evacuation path from hazardous regions.

2.7 Communication Architecture

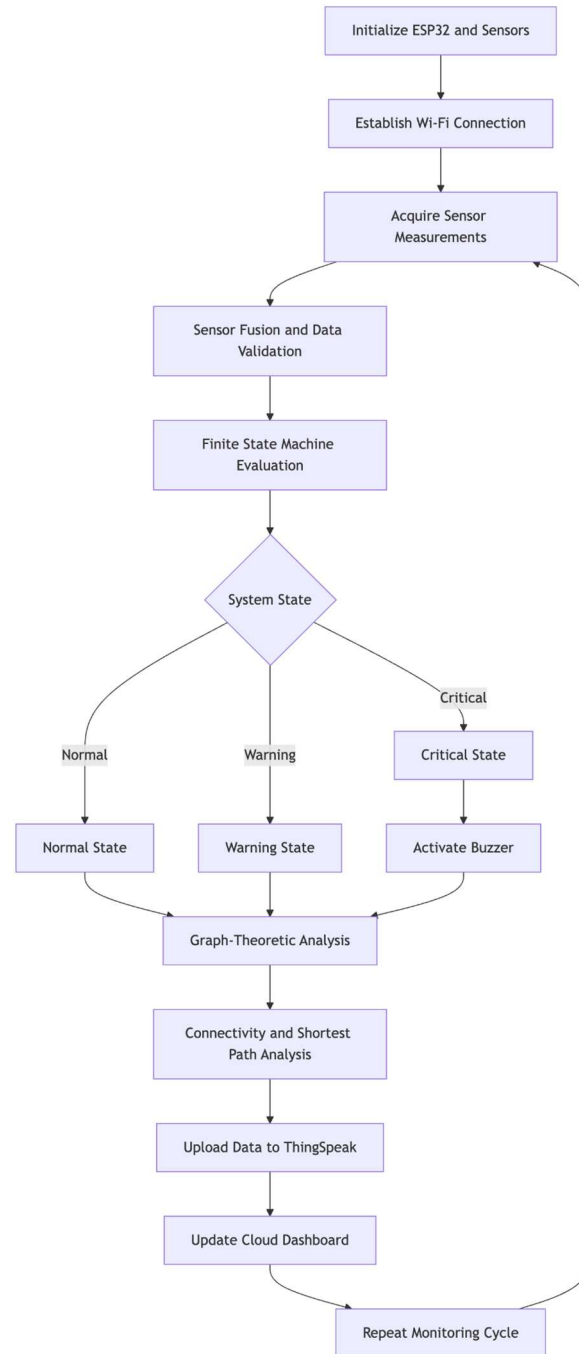


The proposed system employs multiple communication protocols.

Protocol	Purpose
MQTT	Communication between ESP32 nodes and Raspberry Pi
HTTP	Uploading data to ThingSpeak
REST API	Historical data retrieval
WebSocket	Real-time dashboard updates

Chapter 3: Implementation Details

3.1 Implementation



The implementation of the proposed IoT-Based Smart Home Monitoring and Intelligent Decision System follows a systematic approach consisting of data acquisition, preprocessing, intelligent decision-making, graph-theoretic analysis, and cloud-based visualization. The overall methodology adopted for the system is illustrated through the following stages.

1. **Sensor Initialization:** The ESP32 controller initializes all connected sensors, including the DHT11 sensor for temperature and humidity monitoring, the MQ-2 sensor for smoke and gas detection, the PIR sensor for

motion sensing, and the LDR for ambient light measurement. Necessary GPIO pins and communication interfaces are configured during initialization.

2. **Network Establishment:** The ESP32 establishes a Wi-Fi connection to enable communication with the ThingSpeak cloud platform and facilitate remote monitoring.
3. **Data Acquisition:** Real-time measurements of temperature, humidity, smoke concentration, motion activity, and ambient light intensity are continuously collected from the sensors. These readings constitute the primary source of information for the system.
4. **Sensor Fusion and Data Validation:** The acquired sensor values are aggregated into a common sensing layer. Invalid readings are filtered and threshold-based preprocessing is performed to ensure reliable operation. The processed values are combined to obtain an overall representation of the environmental condition.
5. **Finite State Machine Evaluation:** The Intelligent Decision Engine employs a Finite State Machine (FSM) to classify the system into different operating states. Based on the processed sensor information, the system determines whether the current condition corresponds to a Normal, Warning, or Critical state.
6. **Alert Generation:** Whenever hazardous conditions are detected and the system enters the Critical state, an audible alert is generated through the buzzer to immediately notify the occupants.
7. **Graph Analysis:** The smart home network is modeled as a directed graph in which sensors, processing units, cloud services, and actuators are represented as nodes. Connectivity analysis, node degree analysis, and shortest-path algorithms are employed to study communication flow and identify important nodes within the network.
8. **Cloud Data Transmission:** The ESP32 transmits sensor readings, system states, and analytical results to the ThingSpeak cloud platform. The cloud service stores the data and updates graphical dashboards to enable real-time visualization and historical analysis.
9. **Continuous Monitoring:** The entire process of sensing, analysis, decision-making, and cloud communication is executed continuously in a loop to ensure uninterrupted operation and real-time monitoring of the smart home environment.

3.2 Coding Practices

The implementation of the proposed system follows several coding practices to improve readability, maintainability, and scalability.

- **Modularity:** Different functionalities such as sensing, communication, decision-making, and visualization are implemented as separate modules, allowing independent development and easier debugging.
- **Readability:** Meaningful variable names, comments, and a well-structured directory hierarchy are adopted to improve code understanding and simplify future modifications.
- **Maintainability:** Algorithmic components, communication services, and visualization modules are organized into separate files to ensure that changes in one module do not affect the entire system.
- **Reusability:** Frequently used functions and utilities are encapsulated into reusable modules, reducing code duplication and improving consistency.
- **Separation of Concerns:** Hardware interfacing, data processing, cloud communication, and frontend visualization are handled independently, resulting in a cleaner architecture.
- **Scalability:** The modular design allows additional sensors, algorithms, and dashboard components to be incorporated with minimal changes to the existing system.
- **Error Handling:** Input validation and fault detection mechanisms are employed to improve reliability and ensure stable operation under abnormal conditions.

Chapter 4: Results and Discussions

4.1 Execution Outcomes

- **Environmental Monitoring:** The developed system successfully monitored temperature, humidity, smoke concentration, motion activity, and ambient light intensity using multiple sensors connected to ESP32 nodes.
- **Real-Time Visualization:** Sensor readings were transmitted to the ThingSpeak cloud platform and displayed through graphical dashboards, enabling continuous remote monitoring.
- **State-Based Decision Making:** The Finite State Machine correctly classified the system into Normal, Warning, and Critical states according to the sensor conditions.
- **Alert Generation:** The buzzer was activated whenever hazardous conditions were detected, thereby providing immediate notification to the occupants.
- **Graph-Based Analysis:** The smart home network was successfully modeled as a directed graph, and graph algorithms were employed to analyze connectivity and shortest communication paths.

4.2 Validation of Algorithm Correctness

- **Sensor Fusion Validation:** The sensor fusion layer was validated by observing the combined effect of multiple sensor readings. The generated risk score reflected the overall environmental condition rather than relying on individual sensor thresholds.
- **FSM Validation:** The correctness of the Finite State Machine was validated by varying the sensor conditions and observing the corresponding state transitions. Safe environmental conditions resulted in the Normal state, moderate abnormalities triggered the Warning state, and hazardous situations caused the system to enter the Critical state. The observed behavior was consistent with the expected transitions, thereby confirming the correctness of the FSM implementation.
- **Graph Algorithm Validation:** The graph representation correctly modeled the communication structure of the smart home network. Connectivity analysis confirmed that all nodes were reachable, while shortest-path analysis successfully identified efficient communication paths between nodes.

4.3 Handling of Special Cases

- **Sensor Failure:** Invalid sensor readings are filtered during preprocessing to prevent incorrect decisions.
- **Temporary Wi-Fi Disconnection:** In the event of network failure, the ESP32 continuously attempts to reconnect to the Wi-Fi network before resuming data transmission.
- **Noise and Fluctuating Readings:** Threshold-based preprocessing and sensor fusion reduce the effect of transient fluctuations.
- **Smoke Detection During Motion:** The decision engine considers multiple sensor readings simultaneously, thereby providing more reliable hazard assessment than independent threshold-based methods.
- **Missing Cloud Connectivity:** Even if cloud communication is temporarily unavailable, local monitoring and alert generation continue to operate.

4.4 Discussions

The integration of IoT sensors, cloud communication, and graph-theoretic analysis enabled the development of a comprehensive smart home monitoring system. The use of sensor fusion and Finite State Machine concepts improved decision-making capabilities compared with traditional threshold-based systems.

The graph representation provided insights into connectivity and communication flow within the network, while the

ThingSpeak cloud platform enabled real-time visualization and historical analysis. Overall, the system demonstrated reliable operation and effectively combined concepts from IoT, discrete mathematics, graph theory, and intelligent monitoring.

4.5 Limitations

Despite its advantages, the proposed system has certain limitations:

- The system relies on predefined threshold values and does not employ machine learning techniques.
- Sensor accuracy may be affected by environmental noise and calibration errors.
- The graph model assumes a fixed network topology.
- Dependence on Wi-Fi connectivity may affect cloud-based monitoring.
- The system has been tested on a limited number of sensors and nodes.

4.6 Potential Improvements

Several enhancements can be incorporated in future versions of the system:

- Integration of machine learning models for predictive hazard detection.
- Deployment of additional sensors to improve monitoring accuracy.
- Support for larger smart-home networks.
- Development of a mobile application for remote access.
- Implementation of advanced graph algorithms such as A* for intelligent path planning.
- Incorporation of voice assistants and smart actuators.

Chapter 5: Conclusion and Future Scope

5.1 Conclusion

The proposed **IoT-Based Smart Home Monitoring and Intelligent Decision System** was successfully designed and implemented to provide continuous monitoring of environmental and safety parameters. The system integrates multiple sensors with an ESP32 controller, cloud connectivity through ThingSpeak, and intelligent analysis techniques to improve home awareness and security.

The use of sensor fusion enabled multiple sensor readings to be considered collectively rather than independently, resulting in more reliable hazard detection. A Finite State Machine was employed to classify the system into different operating states, thereby facilitating state-based decision making. In addition, graph-theoretic concepts were utilized to model the smart-home network and analyze connectivity and communication paths.

Experimental observations demonstrated that the system was capable of monitoring environmental conditions in real time, generating alerts during abnormal situations, and providing cloud-based visualization and historical data analysis. The modular architecture adopted in the implementation also improves scalability, maintainability, and ease of future expansion.

Overall, the proposed solution effectively combines concepts from the Internet of Things, Computer Networks, Graph Theory, and Discrete Mathematics to develop an intelligent and reliable smart home monitoring system.

5.2 Real-World Applications and Relevance

The developed system has potential applications in several domains, including:

- Smart homes and residential safety systems.
- Smart buildings and intelligent infrastructure.
- Industrial safety and environmental monitoring.
- Remote monitoring systems for elderly care and healthcare facilities.
- IoT-based research and educational applications.

The modular design and cloud-based monitoring capabilities make the system suitable for both small-scale and large-scale deployments.

5.3 Future Scope

Although the developed system performs effectively, several enhancements can be incorporated in future versions:

- Integration of machine learning techniques for predictive hazard detection.
- Support for additional sensors to improve monitoring accuracy.
- Development of a dedicated mobile application for remote access and notifications.
- Implementation of advanced graph algorithms for intelligent evacuation and path optimization.
- Incorporation of voice assistants and smart actuators for automated responses.
- Deployment of the system in larger smart buildings and industrial environments.

These improvements can further enhance the intelligence, reliability, and scalability of the proposed system and extend its applicability to a wider range of real-world scenarios.

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